customer Performance Analysis

1. Introduction:

This report presents a comprehensive analysis of customer behavior based on transactional data. The objective is to identify patterns in purchasing behavior, customer segmentation, churn trends, and revenue generation to support data-driven business decisions

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

## Load the data

df = pd.read_csv("C:/Users/Pratiksha
Swami/Downloads/archive/ecommerce_customer_data_custom_ratios.csv")
df.head()
```

```
   Customer ID        Purchase Date Product Category  Product Price
Quantity  \
0        46251  2020-09-08 09:38:32      Electronics             12
3
1        46251  2022-03-05 12:56:35             Home            468
4
2        46251  2022-05-23 18:18:01             Home            288
2
3        46251  2020-11-12 13:13:29         Clothing            196
1
4        13593  2020-11-27 17:55:11             Home            449
1

   Total Purchase Amount Payment Method  Customer Age  Returns  \
0                    740    Credit Card            37      0.0
1                   2739         PayPal            37      0.0
2                   3196         PayPal            37      0.0
3                   3509         PayPal            37      0.0
4                   3452    Credit Card            49      0.0

         Customer Name  Age  Gender  Churn
0  Christine Hernandez   37    Male      0
1  Christine Hernandez   37    Male      0
2  Christine Hernandez   37    Male      0
3  Christine Hernandez   37    Male      0
4          James Grant   49  Female      1
```

2. Data Cleaning & Preprocessing

Data cleaning was performed to ensure accuracy and consistency. Missing values in the Returns column were handled by replacing them with zero. The Purchase Date column was converted into datetime format for time-based analysis. Feature engineering techniques were applied, including extraction of Month from Purchase Date and validation of Total Purchase Amount. Duplicate or inconsistent records were checked and handled appropriately Data Cleaning

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
 #   Column                  Non-Null Count   Dtype
---  ------                  --------------   -----
 0   Customer ID             250000 non-null  int64
 1   Purchase Date           250000 non-null  object
 2   Product Category        250000 non-null  object
 3   Product Price           250000 non-null  int64
 4   Quantity                250000 non-null  int64
 5   Total Purchase Amount   250000 non-null  int64
 6   Payment Method          250000 non-null  object
 7   Customer Age            250000 non-null  int64
 8   Returns                 202404 non-null  float64
 9   Customer Name           250000 non-null  object
 10  Age                     250000 non-null  int64
 11  Gender                  250000 non-null  object
 12  Churn                   250000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB

df['Purchase Date']=pd.to_datetime(df['Purchase Date'])

df.isnull().sum()

Customer ID               0
Purchase Date             0
Product Category          0
Product Price             0
Quantity                  0
Total Purchase Amount     0
Payment Method            0
Customer Age              0
Returns                   0
Customer Name             0
Age                       0
Gender                    0
Churn                     0
dtype: int64
```

```
df['Returns'] = df['Returns'].fillna(0)

df.drop(columns=['Age'],inplace=True)

df.describe()
```

```
        Customer ID                   Purchase Date  Product Price  \
count  250000.00000                          250000  250000.000000
mean    25004.03624  2021-11-06 23:31:24.372304384     254.659512
min         1.00000            2020-01-01 00:15:00      10.000000
25%     12497.75000  2020-12-02 19:33:23.249999872     132.000000
50%     25018.00000            2021-11-06 13:10:59     255.000000
75%     37506.00000  2022-10-11 03:42:32.750000128     377.000000
max     50000.00000            2023-09-15 12:24:08     500.000000
std     14428.27959                            NaN     141.568577

            Quantity  Total Purchase Amount   Customer Age
Returns  \
count  250000.000000          250000.000000  250000.000000
250000.000000
mean        2.998896            2725.370732      43.940528
0.403076
min         1.000000             100.000000      18.000000
0.000000
25%         2.000000            1477.000000      31.000000
0.000000
50%         3.000000            2724.000000      44.000000
0.000000
75%         4.000000            3974.000000      57.000000
1.000000
max         5.000000            5350.000000      70.000000
1.000000
std         1.414694            1442.933565      15.350246
0.490517

               Churn
count  250000.000000
mean        0.199496
min         0.000000
25%         0.000000
50%         0.000000
75%         0.000000
max         1.000000
std         0.399622
```

3. Exploratory Data Analysis (EDA)

EDA was conducted to understand customer behavior and business trends. Various visualizations were used to uncover insights:

- Product Category Analysis: Identified top-performing categories contributing to revenue.
- Monthly Sales Trend: Revealed seasonal patterns and fluctuations in sales.
- Churn Analysis: Highlighted customer retention challenges.
- Age Distribution: Showed the demographic profile of customers.
- Payment Method Analysis: Identified preferred transaction methods

```
df['Calculated_Total'] = df['Product Price'] * df['Quantity']

# Check mismatch
mismatch = (df['Calculated_Total'] != df['Total Purchase
Amount']).sum()
print("Mismatch rows:", mismatch)
```

```
Mismatch rows: 249952
```

"The dataset contained a precomputed 'Total Purchase Amount' which significantly differed from the calculated value (Price × Quantity). This indicates the presence of additional business factors such as discounts, taxes, or custom pricing logic. Therefore, both metrics were retained for analysis."

```
df['Difference'] = df['Total Purchase Amount'] -
df['Calculated_Total']
```

RFM Analysis , r for recency : how recently they purchased ,f for frequency : how frquently customer purchase and M for monetary like how much customer spend

```
import datetime as dt

today = df['Purchase Date'].max()

rfm = df.groupby('Customer ID').agg({
    'Purchase Date': lambda x: (today - x.max()).days,
    'Customer ID': 'count',
    'Total Purchase Amount': 'sum'
})

# rename the columns
rfm.columns = ['Recency', 'Frequency', 'Monetary']
rfm.head()
```

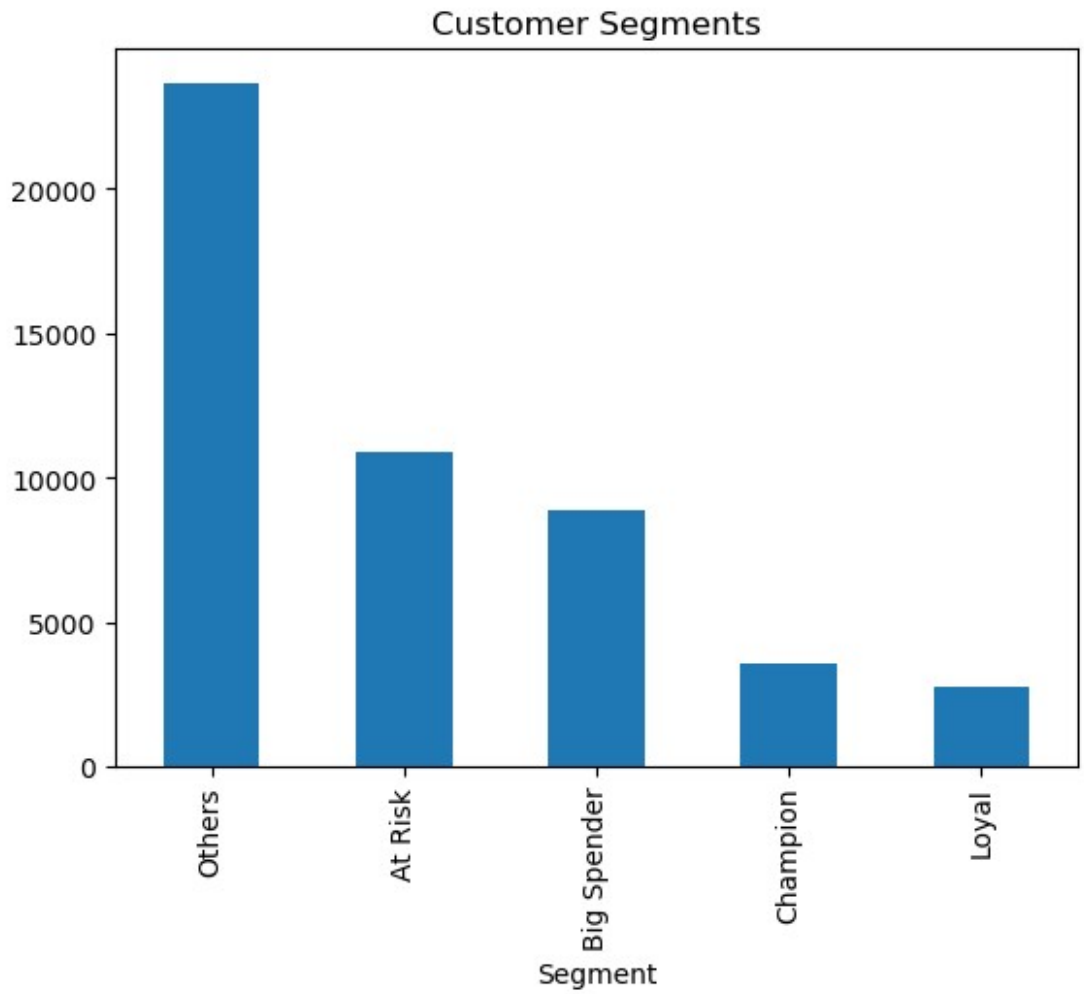
```
             Recency  Frequency  Monetary
Customer ID
1                 57          1      3491
2                298          3      7988
3                 88          8     22587
4                126          4      8715
5                170          8     12524
```

```
rfm['R_score'] = pd.qcut(rfm['Recency'], 4, labels=[4,3,2,1])
rfm['F_score'] = pd.qcut(rfm['Frequency'], 4, labels=[1,2,3,4])
rfm['M_score'] = pd.qcut(rfm['Monetary'], 4, labels=[1,2,3,4])

rfm['RFM_Score'] = (
    rfm['R_score'].astype(str) +
    rfm['F_score'].astype(str) +
    rfm['M_score'].astype(str)
)

rfm
```

```
             Recency  Frequency  Monetary R_score F_score M_score
RFM_Score
Customer ID

1                 57          1      3491       4       1       1
411
2                298          3      7988       2       1       1
211
3                 88          8     22587       3       4       4
344
4                126          4      8715       3       2       1
321
5                170          8     12524       3       4       2
342
...              ...        ...       ...     ...     ...     ...
...
49996            271          4     14044       2       2       3
223
49997             49          8     22431       4       4       4
444
49998             11          4      8610       4       2       1
421
49999            419          2      6984       1       1       1
111
50000            126          2      4116       3       1       1
311

[49673 rows x 7 columns]
```

```
def segment(row):
    if row['RFM_Score'] == '444':
        return 'Champion'
    elif row['M_score'] == 4:
        return 'Big Spender'
    elif row['F_score'] == 4:
        return 'Loyal'
    elif row['R_score'] == 1:
        return 'At Risk'
```

```
    else:
        return 'Others'

rfm['Segment'] = rfm.apply(segment, axis=1)

rfm['Segment'].value_counts().plot(kind='bar')
plt.title("Customer Segments")
plt.show()
```



Champion : A small group of customers contributes most revenue,

loyal:Some customers are frequent but low spenders,

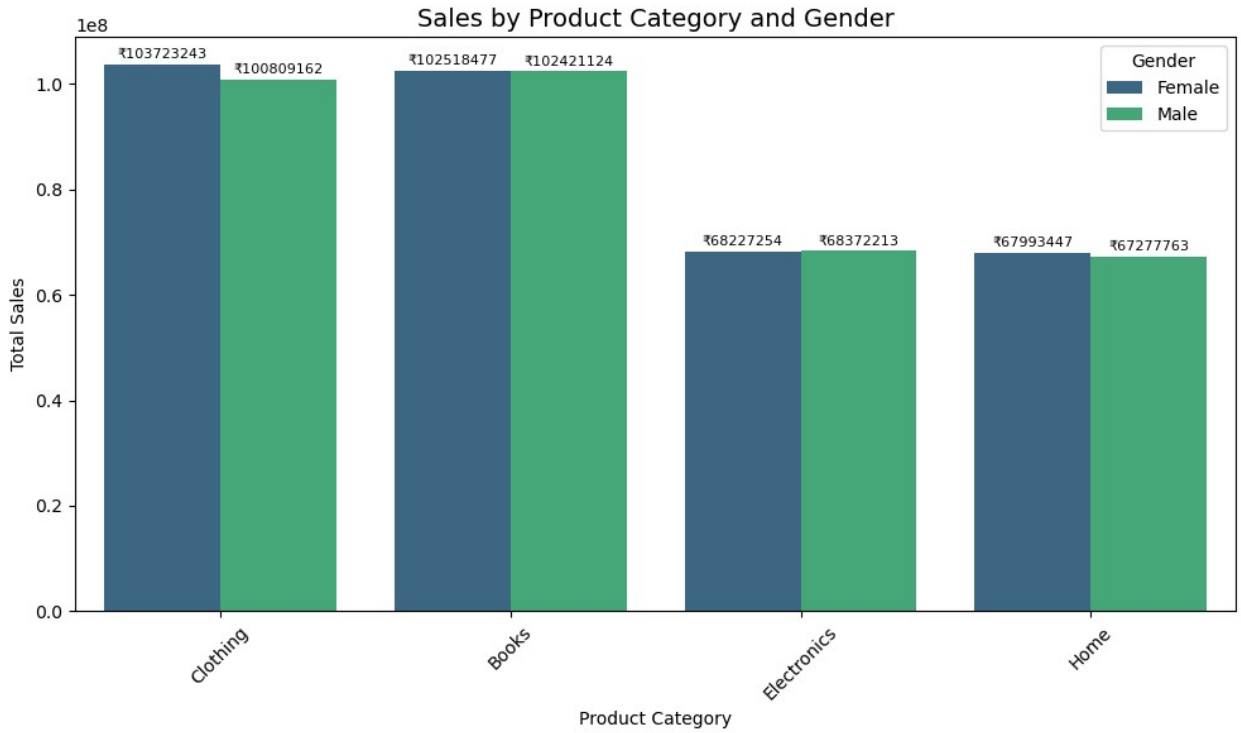
at risk :Many customers are inactive → need retention,

big Spender : High-value customers should be targeted

“These customer segments help businesses design targeted marketing strategies, improve customer retention, and increase overall revenue.”

4 Visualization

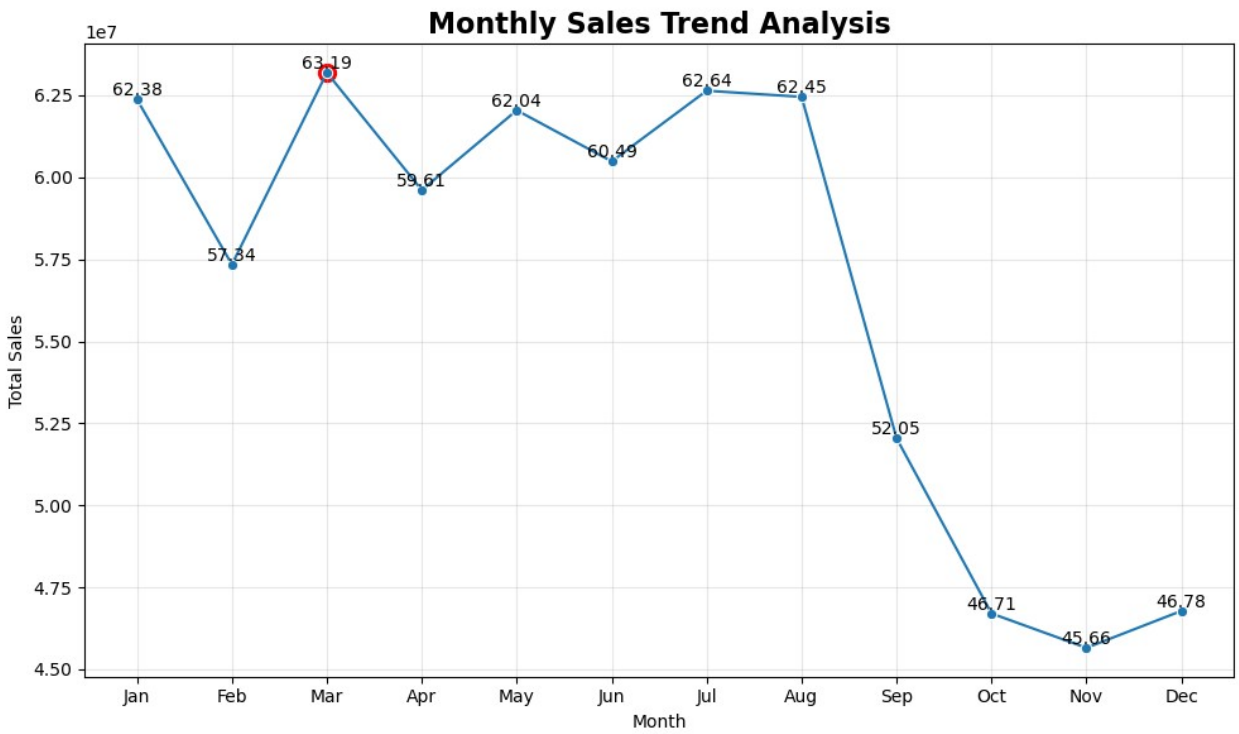
```
# 1. Sales by Product Category

Category_Sale = df.groupby(['Product Category','Gender'])['Total
Purchase Amount'] \
    .sum().reset_index()

Category_Sale = Category_Sale.sort_values(by='Total Purchase Amount',
ascending=False)

plt.figure(figsize=(10,6))

ax = sns.barplot(
    data=Category_Sale,x='Product Category',y='Total Purchase
Amount',hue='Gender',palette='viridis'
)

# Add labels
for container in ax.containers:
    ax.bar_label(container, fmt='₹%.0f', padding=2, fontsize=8)

plt.title("Sales by Product Category and Gender", fontsize=14)
plt.xlabel("Product Category")
plt.ylabel("Total Sales")

plt.xticks(rotation=45)  # IMPORTANT for readability

plt.tight_layout()
plt.show()
```



The visualization shows sales distribution across product categories segmented by gender, highlighting differences in purchasing behavior and preferences.

```
# 2 Monthly Sales Trend

df['Month'] = df['Purchase Date'].dt.month

monthly_sales = df.groupby('Month')['Total Purchase
Amount'].sum().reset_index()
import calendar
monthly_sales['Month'] = monthly_sales['Month'].apply(lambda x:
calendar.month_abbr[x])

plt.figure(figsize=(10,6))

ax = sns.lineplot(data=monthly_sales,x='Month',y='Total Purchase
Amount',marker='o')

# Add data labels
for x, y in zip(monthly_sales['Month'], monthly_sales['Total Purchase
Amount']):
    plt.text(x, y, f'{y/1e6:.2f}', ha='center', va='bottom')
#highlight peak month
max_row = monthly_sales.loc[monthly_sales['Total Purchase
Amount'].idxmax()]

plt.scatter(max_row['Month'], max_row['Total Purchase Amount'],
color='red', s=100)
```

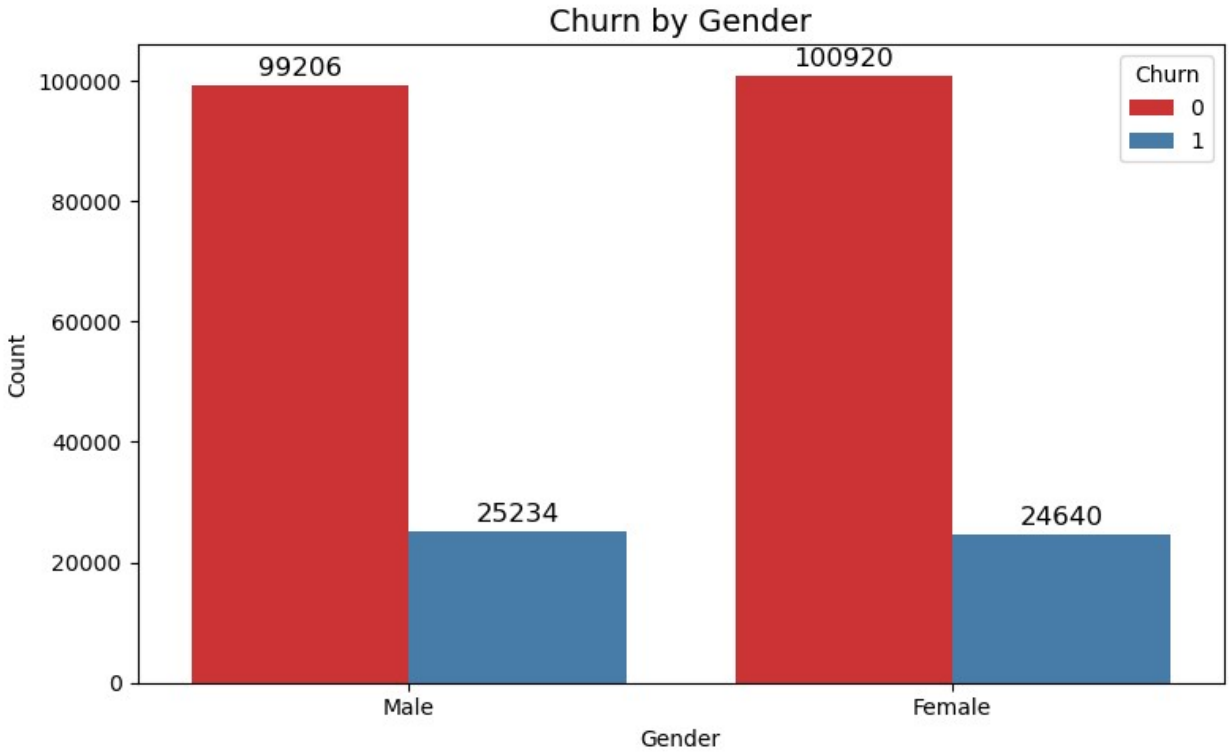
```
plt.title("Monthly Sales Trend Analysis",
fontsize=16,fontweight='bold')
plt.xlabel("Month")
plt.ylabel("Total Sales")

plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



The monthly sales trend shows strong and consistent performance from January to August, with peak sales observed in March and July. However, a significant decline is observed from September to November, indicating potential seasonal effects or reduced customer engagement. A slight recovery in December suggests the impact of end-of-year promotions or festive demand.

```
# 3 Customer Churn Analysis

plt.figure(figsize=(8,5))

ax = sns.countplot(data=df,x='Gender',hue='Churn',palette='Set1')

for container in ax.containers:
    ax.bar_label(container, padding=2, fontsize=12)

plt.title("Churn by Gender", fontsize=14)
```

```
plt.xlabel("Gender")
plt.ylabel("Count")

plt.tight_layout()
plt.show()
```

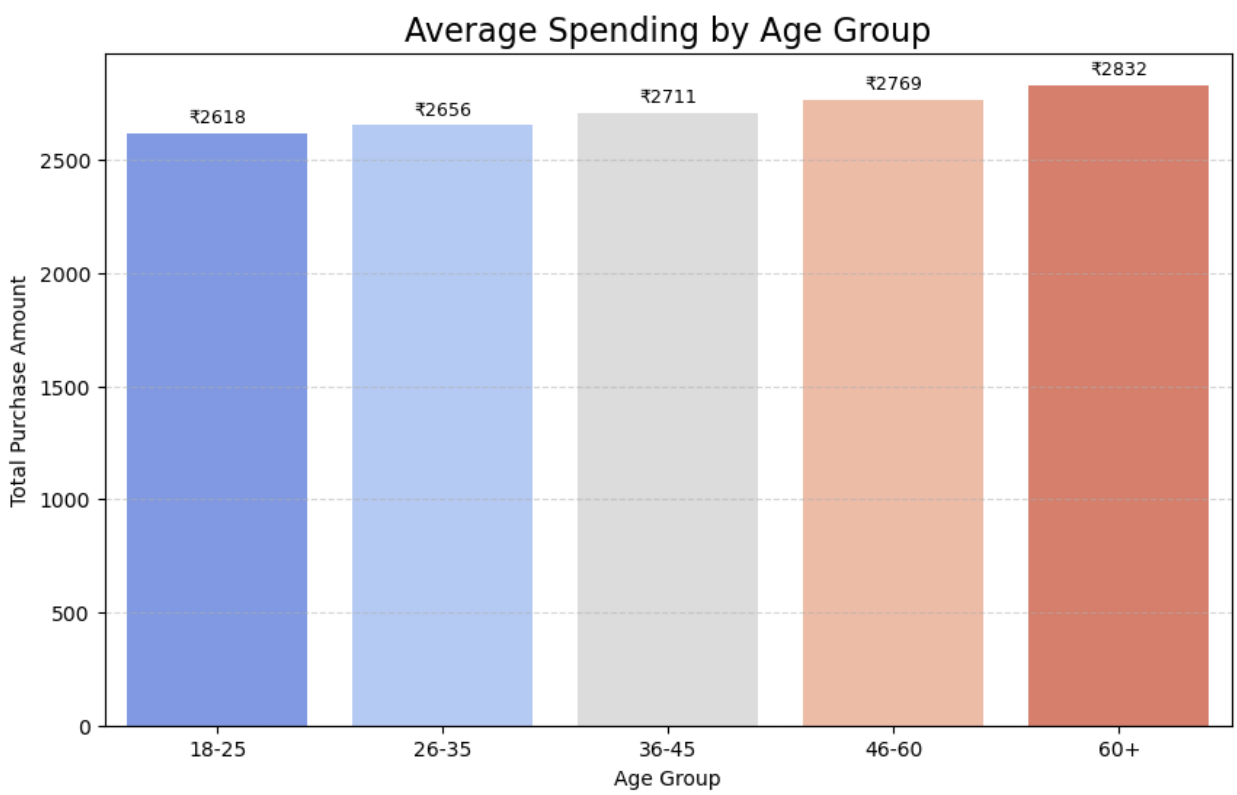


Churn behavior varies across gender, indicating differences in customer retention patterns.

```
#4 Customer Age Distribution

# Create age groups
df['Age Group'] = pd.cut(df['Customer Age'], bins=[18,25,35,45,60,80],
                         labels=['18-25','26-35','36-45','46-
60','60+'])

age_sales = df.groupby('Age Group')['Total Purchase
Amount'].mean().reset_index()

plt.figure(figsize=(10,6))

ax = sns.barplot(data=age_sales,x='Age Group',y='Total Purchase
Amount',palette='coolwarm')

for container in ax.containers:
    ax.bar_label(container, fmt='₹%.0f', padding=3, fontsize=9)
```

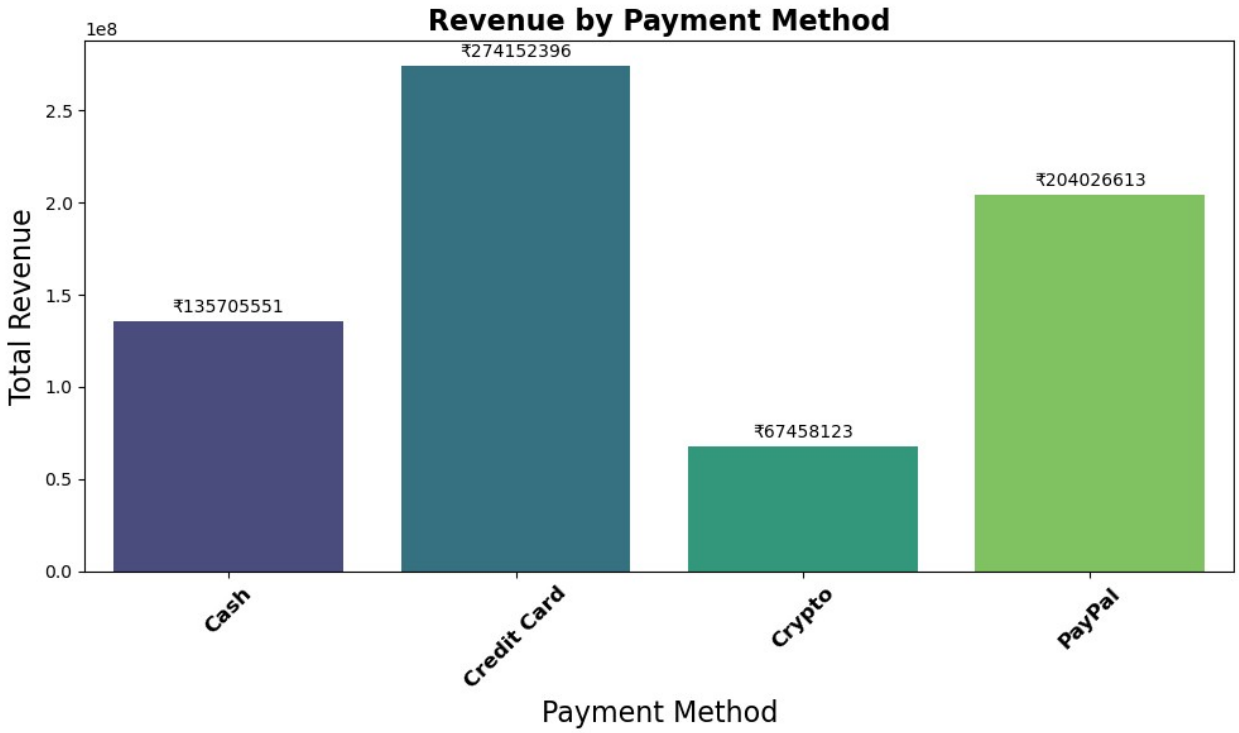
```
plt.title("Average Spending by Age Group", fontsize=16)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()
```

```
C:\Users\Pratiksha Swami\AppData\Local\Temp\
ipykernel_9660\293302280.py:5: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future
version of pandas. Pass observed=False to retain current behavior or
observed=True to adopt the future default and silence this warning.
  age_sales = df.groupby('Age Group')['Total Purchase
Amount'].mean().reset_index()
C:\Users\Pratiksha Swami\AppData\Local\Temp\
ipykernel_9660\293302280.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  ax = sns.barplot(data=age_sales,x='Age Group',y='Total Purchase
Amount',palette='coolwarm')
```



The analysis shows that average spending increases with age. Customers aged 60+ contribute the highest spending, while younger customers (18–25) spend comparatively less. This indicates that older customers have higher purchasing power and represent a valuable target segment.

```python
# 5 Payment Method Analysis

plt.figure(figsize=(10,6))

payment_sales = df.groupby('Payment Method')['Total Purchase
Amount'].sum().reset_index()

ax = sns.barplot(
    data=payment_sales,
    x='Payment Method',
    y='Total Purchase Amount',
    hue='Payment Method',
    palette='viridis',
    legend=False
)

# Labels
for container in ax.containers:
    ax.bar_label(container, fmt='₹%.0f', padding=3)

plt.title("Revenue by Payment Method", fontsize=16, fontweight='bold')
plt.xlabel("Payment Method",fontsize=16)
plt.ylabel("Total Revenue",fontsize=16)

plt.xticks(rotation=45,fontsize=12,fontweight='bold')

plt.tight_layout()
plt.show()
```



The analysis shows that credit card payments contribute the highest revenue,followed by Paypal . Cash transaction are moderate , while cryptocurrency is the least used payment method.This indicates a strong preference for digital and card-based payments among customers

5 RECOMMENDATIONS

1. Focus on High-Value Customers

Customers with high spending (RFM analysis) contribute the most revenue. The company should provide loyalty programs, exclusive offers, and premium services to retain these valuable customers.

1. Improve Customer Retention (Reduce Churn)

A noticeable number of customers have churned. The company should implement retention strategies such as personalized offers, email campaigns, and discounts to re-engage inactive customers.

1. Target Older Age Groups for Revenue Growth

Customers aged 46+ show higher spending behavior. Marketing efforts should focus on this segment with premium products and personalized experiences.

1. Encourage Digital Payment Methods

Credit cards and PayPal generate the highest revenue. The company should promote these methods through cashback offers, discounts, and seamless payment experiences.

1. Improve Low-Performing Areas

Certain product categories and payment methods (like crypto) show lower performance. The company should analyze these areas and improve offerings or customer experience.

1. Leverage Seasonal Trends

Sales drop significantly after August. The company should introduce seasonal campaigns, festive offers, and promotions during low-performing months to boost sales.